Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CIC52]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

DATA COMMUNICATION AND COMPUTER NETWORKS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Draw OSI reference model with and explain functions of each layers.	**10**
	b.	Explain data communication model with a neat diagram.	**04**
2.	**a.**	List and explain the categories of line coding scheme in digital to digital conversion.	**10**
	b.	Define the following: **i.** Baseline wandering **ii.** Bandwidth **iii.** DC components **iv.** Self synchronization	**04**
3.	**a.**	Explain how Dijkstra's algorithm is used to find the least cost route with a help of a suitable example.	**10**
	b.	What is Multiplexing? List the categories of multiplexing	**04**
		OR	
4.	**a.**	Differentiate circuit switched and packet switched networks with suitable diagram	**10**
	b.	Differentiate between synhronous TDM and statistical TDM.	**04**
5.	**a.**	Outline the key elemets of LAN.	**10**
	b.	Write a note on Traffic management.	**04**
		OR	
6.	**a.**	Write a note on layer 2 and layer 3 switches	**10**
	b.	Differentiate between switches and routers.	**04**
7.	**a.**	Discuss the various mechanisms used for congestion control.	**10**
	b.	Write a note on Spanning tree protocol	**04**
		OR	
8.	**a.**	Briefly explain the connection oriented transport protocol mechanisms.	**10**
	b.	Describe all the fields of IPv4 header.	**04**